

Lab: Probability and Generative Models

Learning Goal: Build intuition for probabilistic reasoning and generative model construction.

Part 1: Bayesian Reasoning Practice

Exercise: For each scenario, identify the prior, likelihood, and compute the posterior using Bayes' theorem.

1. **Medical test:** A disease affects 1% of the population (prior). A test has 95% sensitivity ($P(\text{positive} \mid \text{disease}) = 0.95$) and 90% specificity ($P(\text{negative} \mid \text{no disease}) = 0.90$). You test positive. What's the probability you actually have the disease?
2. **Email spam:** 40% of your emails are spam (prior). $P(\text{contains "free"} \mid \text{spam}) = 0.8$, $P(\text{contains "free"} \mid \text{not spam}) = 0.1$. An email contains the word "free." What's the probability it's spam?

Show your calculations step by step.

Write your response here

Part 2: Building Generative Models

Learning Goal: Construct a simple generative model from scratch.

Exercise: Build a generative model for weather and clothing:

- Hidden state $s \in \{\text{sunny, rainy}\}$
- Observation $o \in \{\text{umbrella, sunglasses, jacket}\}$

- Define $P(s)$ — the prior on weather states
- Define $P(o | s)$ — the likelihood of each observation given each weather state
- Verify: Does your model make intuitive sense? ($P(\text{umbrella} | \text{rainy})$ should be high)
- Use Bayes' theorem: If you observe someone carrying an umbrella, what's the posterior $P(s | \text{umbrella})$?

Write your response here

Part 3: Hidden Markov Model

Learning Goal: Extend static models to dynamic sequences.

Scenario: A simple weather HMM:

- States: {sunny, rainy}
- Transition: $P(\text{sunny} \rightarrow \text{sunny}) = 0.7$, $P(\text{sunny} \rightarrow \text{rainy}) = 0.3$, $P(\text{rainy} \rightarrow \text{rainy}) = 0.6$, $P(\text{rainy} \rightarrow \text{sunny}) = 0.4$
- Observation: $P(\text{umbrella} | \text{rainy}) = 0.9$, $P(\text{umbrella} | \text{sunny}) = 0.2$

Day 1: You observe an umbrella. Day 2: You observe sunglasses. Day 3: You observe an umbrella.

1. What's the most likely weather sequence?
2. How does the inference at each step depend on the previous day?

Write your response here

Part 4: Connecting Math to Concepts

Learning Goal: Bridge mathematical formalism to cognitive science concepts.

Exercise: Map each mathematical object to its cognitive science equivalent:

Math	Cognitive Science
$P(s)$ — prior	
$P(o \mid s)$ — likelihood	
$P(s \mid o)$ — posterior	
s — hidden state	
o — observation	
$P(s_t \mid s_{\{t-1\}})$ — transition	

Write your response here

Part 5: Reflection

In 150 words, reflect: Why is it useful to formalize intuitive concepts (like "belief" and "surprise") in mathematical language? What do we gain and what do we lose?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Bayesian computation	$\text{Prior} \times \text{Likelihood} \rightarrow \text{Posterior}$
2	Model building	Generative model construction
3	Temporal reasoning	Hidden Markov Models
4	Concept mapping	$\text{Math} \leftrightarrow \text{Cognitive science bridge}$
5	Epistemological reflection	Value of formalization